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A Novel Use of Normally Autonomous Subsea Drones in Direct ROV Mode for Shallow Water Pipeline Surveys

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Abstract

This paper describes an innovative method of undertaking pipeline and seabed surveys by adapting an otherwise autonomous underwater inspection drone (UID) to operate in a tether-less human-operated ROV mode which can be launched and recovered and operated remotely at a long distance from a lay vessel, or small support vessel, without need for a large Multi-role Support Vessel (MSV).

A normally-autonomous subsea drone is adapted by using a surface buoy with a small data cable to the vehicle and with telemetry to enable direct control with real-time data and video feed to a nearby vessel or a remote control centre via Wi-Fi, cellular or satellite link.

The vehicle is fitted with its standard integral onboard survey sensors such as Multi-Beam Echosounder Sonar, dual *CathX Ocean* UHD camera system, enabling geolocation and 3D point-cloud reconstruction and metrology if required, multi-sensor leak detection, optical automatic pipe-tracking and Inertial Navigation System, hydrography sensors; USBL acoustic position for independent positional information.

The advantages of operating the UID in human-operated ROV mode over a classic ROV are:

- Disconnects the vehicle from the offshore platform or support vessel. Meaning that the vehicle can be operated in direct human operated ROV mode from long distance by way of Wi-Fi, 4G/5G cellular network or by satellite comms from the buoy.
- The surface buoy, in conjunction with USBL transponders on both the vehicle and the buoy enables independent visual and acoustic tracking from a nearby vessel, thus alleviating any security concerns.
- The dual UHD camera system (by CathX Ocean) enables extremely high-definition video quality as well as photogrammetric acquisition of geo-referenced point-cloud data, thus enabling highly accurate metrology and 3D model reconstruction.
- The built-in MBES system in combination with the point-cloud video data can be overlaid to provide a multi-source seabed and pipeline/structure geo-referenced image.
- The autonomous nature of the drone can allow it to automatically acquire and maintain track of the target pipeline or structure, reducing the need for skilled human navigation.

- Provides a technological and regulatory pathway to eventual full autonomous operations if desired.
- Can operate as a "force multiplier" by allowing parallel working in an area with say a conventional ROV or USV.

The adaptation for use of UID/AUV in ROV mode combines many of the technical advantages of the autonomous drone such as significantly better video and data acquisition as well as the elimination of large support vessel

Is to be in close proximity, with the advantages of real-time control and video / data acquisition, whilst also addressing many security concerns related to remote underwater vehicles.

Introduction

Asset integrity management is a continuous, cyclical, and iterative process aimed at ensuring the safe, cost-effective, and reliable operation of subsea production and pipeline systems. This is achieved through regular inspections of offshore subsea assets, typically involving visual assessments, cathodic protection measurements, and pipeline profiling.

Traditionally, the oil and gas industry has relied on dedicated vessels equipped with one or two Remotely Operated Vehicles (ROVs) and the necessary tools to carry out these inspections.

However, the landscape is evolving. As subsea fields adopt more advanced technologies to boost production, the need for higher safety standards and more frequent inspections has grown. At the same time, operators are under pressure to reduce operational expenditures (OPEX) while maintaining high system availability.

In this context, fully remotely operated or even autonomous vehicles are proving to be transformative. Underwater Intervention/Inspection Drones (UIDs) offer a timely and cost-effective alternative to conventional Inspection, Maintenance, and Repair (IMR) services, eliminating the need for surface vessels. These drones are also crucial for emergency response, enabling rapid environmental monitoring and inspection, and significantly reducing downtime in subsea production.

A notable example of such innovation is Saipem's FlatFish—a hybrid subsea robotic system that combines features of both ROVs and Autonomous Underwater Vehicles (AUVs). Designed for efficient subsea inspections, the UID minimizes intervention time and simplifies logistics related to deployment and recovery. Originally developed as a research initiative, it has undergone extensive validation and qualification, culminating in a fully industrialized product. Its capabilities were successfully demonstrated in a deepwater pilot test offshore Brazil at a depth of 1,750 meters.



Figure 1—Hydroner-R™ (on the left), FlatFish UID (on the right)

This paper describes a method of operation that combines the advantages of fully autonomous UID with the conventional direct human operated ROV to enable safer, faster, higher quality and more efficient inspections and survey of subsea pipelines and other subsea assets.

A distinction is made here between an Autonomous Underwater Vehicle (AUV) and an Underwater Inspection (or Intervention) Drone (UID): whereas AUV can be considered a general term for non-human operated subsea vehicles, a UID, on the other hand, can be considered as an AUV which is capable of ROV-like movement (hovering, vertical and lateral). This paper describes the adaptation of such a UID to also be human operated in so-called *ROV mode*.

Autonomous inspection of Subsea live assets

The rapid advancement of technology and the evolution of Artificial Intelligence have laid the foundation for significantly enhanced capabilities in robotic systems across various industries. In subsea robotics, recent innovations have introduced a new generation of autonomous features that were previously unimaginable in earlier underwater inspection vehicles.

Today, advanced hybrid Autonomous Underwater Vehicles (AUVs) are capable of performing complex inspection tasks across both shallow and deepwater offshore energy fields. These capabilities mark a major technological leap, characterized by several key features:

Hovering Capability: Modern hybrid AUVs now possess precise navigation control systems that allow them to hover around fixed structures. This enables the collection of general visual data or point clouds while maintaining a safe distance. The vehicle can smoothly maneuver, pause in front of specific components, and capture close visual inspection (CVI) data—something that traditional AUVs were not capable of just a few years ago.

Vertical Navigation: Some hybrid AUVs can autonomously navigate along the vertical axis of the water column. This allows for detailed inspection of vertical structures like risers and mooring lines. The vehicle maintains a predefined distance from the asset while capturing stereo images and video. Using onboard sensors such as mechanical and imaging sonar, it can track and inspect structures even when their exact shape is not known in advance.

Landmark-Based Navigation: The ability to orient and localize itself in relation to its environment is a critical advancement for autonomous operations. Inspired by self-driving car technologies, subsea vehicles now incorporate deep learning algorithms for object recognition. These algorithms enable the AUV to distinguish between different subsea structures (e.g., pipelines vs. umbilicals, or Christmas trees vs. manifolds), identify specific components like ROV panels or valves, and use these landmarks to refine its positioning and inspection path.

Autonomous Anomaly Detection: Building on its recognition capabilities and access to extensive field data, the hybrid AUV can autonomously detect anomalies in real time (online) or during post-mission analysis (offline). This allows the system to identify unusual or unexpected conditions around subsea assets, helping operators focus on potential issues early and prevent critical failures.

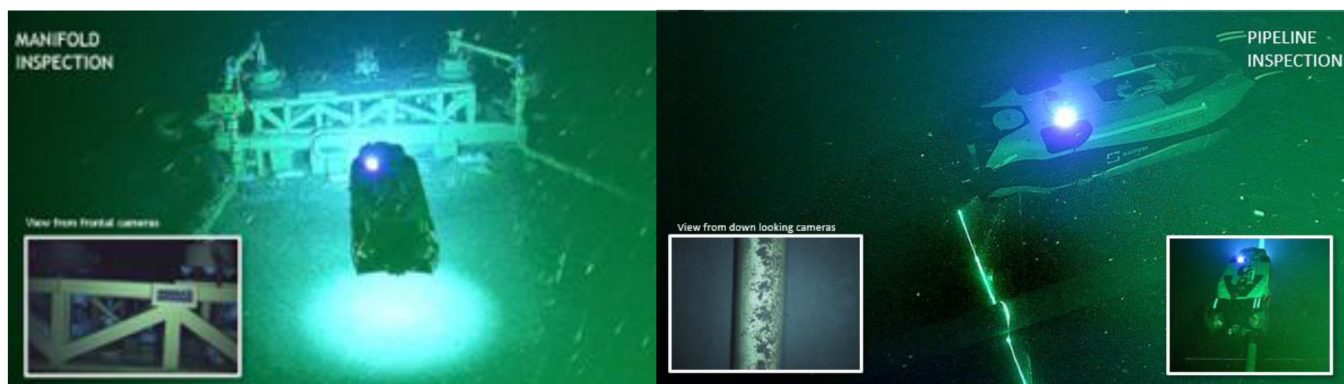


Figure 2—UID performing autonomous inspection of fixed structures (on the left), UID performing autonomous tracking of a pipeline (on the right)

Subsea operations methods by Underwater Inspection Drones

Among the new features and capabilities embedded into the new generation of UID, several operational modes have been introduced. In fact, an UID allows great flexibility in deployment and operating modes.

Deployment modes

- **Onshore Itinerant**

Launched and recovered from the quayside using a crane with a remote detachable hook, allowing to avoid an expensive vessel with offshore crane for supporting the inspection campaign. This is more adapted to shallow water and nearshore applications, but recent evaluations have led to extending this solution to other scenarios with longer distances.

- **Offshore Itinerant**

Launched and recovered from a platform or a support vessel by way of:

- A crane with a remote detachable hook, or
- By way of a "Floating Garage" lowered to the sea surface and from which the UID can then exit and enter without risk of wave action.
- Leveraging existing subsea winches placed on platform or FPSO: for surface assets equipped with a subsea winch, the UID can be deployed utilizing a suppressor (deadweight) to reach the seabed water level
- Utilizing existing surface crane deploying the vehicle, potentially within its Floating or Flying Garage, can be lowered to a shallow water depth. From this point, the vehicle can autonomously navigate through the water column to reach the sea bottom and assets to be inspected.

- **Offshore Resident**

The UID remains underwater for long dives, such as 6 to 12 months, through a semi-permanent garage or a mobile "Flying Garage" positioned on the seabed allowing the UID to be recharged and transfer data and new mission instructions by way of an umbilical to the surface. In this case, after the vehicle mobilization, no support vessel is required for the entire dive duration reducing offshore personnel, costs and CO2 emissions.

The subsea docking station umbilical can either be routed to a platform-based control room or attached to a floating buoy with self-contained power and communications, such as the *HyBuoy*, thus allowing data and missions to be sent wirelessly to a remote-control room.

Operating modes

- **Fully Autonomous mode**

The UID is programmed with the mission and will proceed to track the pipeline or other assets collecting video and other sensor data. Upon completion of the planned mission the UID will self-locate to the programmed collection location for recovery or, in the case of resident deployment, to the garage for recharging and data transfer.

- **Supervised AUV mode**

While operating in autonomous mode, the UID can be monitored, and data can be transferred to a control room including real-time video and other sensor data. Additionally, an operator can intervene and temporarily assume direct control of UID if desired.

Real-time data and video communications can be achieved by several channels:

- By way of a surface "**communications buoy**" which is towed by the UID with a small diameter umbilical. The communications buoy is fitted with GPS receiver and a telemetry link using long-range WiFi, 5G cellular or satellite. Additionally, the communications buoy is fitted with a transponder to enable USBL tracking of UID.
- Within close range of a platform or other structure, it is possible to achieve real-time video, data and control communications by way of an **optical modem**. UIDs are fitted with an in-built optical modem for this purpose.
- Within several kilometers it is also possible to transmit instructions and receive limited data by way of an in-built **acoustic modem**, depending on the water depth

- **ROV mode**

The UID can be operated in direct human operated ROV mode:

- In *Tethered ROV mode* the UID is connected directly to the support vessel with a data/video umbilical allowing it to be operated exactly as a conventional ROV. This mode may be useful for certain limited range activities which do not require the vessel to move away from the work location.
- In *Untethered ROV mode*, communication with the UID is achieved by one of the channels described above in Supervised mode. The difference being that the UID is continuously controlled by the ROV pilot. The vessel is then free to move far away from the work-site and conduct other parallel activities.

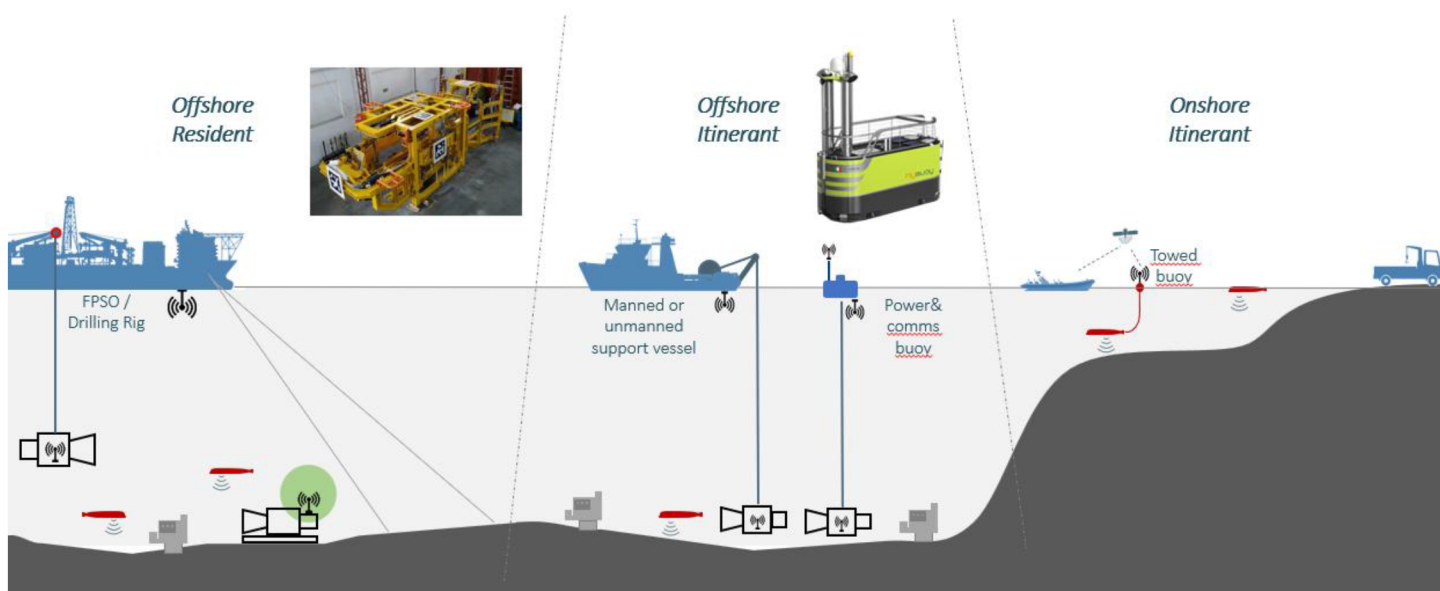


Figure 3—Subsea drones deployment modes in different application scenarios

ROV mode of operation for pipeline survey with Underwater Inspection Drones

The subject of this paper describes the experience and the realized benefits of *Offshore Itinerant* deployment of the UID using a vessel and crane with a detachable hook to operate in both *Tethered* and *Untethered ROV mode* using the towed *Communication Buoy* and Wi-Fi telemetry to the support vessel. It is believed that this is the first-time use of such a method of inspection based on the novel adaptation of a normally autonomous subsea drone.

This method of operation was used for the as-laid survey of subsea pipelines in relatively shallow water with Multibeam Echo Sounding (MBES) and UHD video capture. It is also possible to add additional sensors to the UID including contactless CP measurement, acoustic and optical leak detection, sub-bottom profiler, pipe-tracker etc. Some of these have been tested in controlled environment and some are technically feasible but not yet tested.

First use of the UID in ROV mode for the as-laid survey of pipeline in shallow water application in Middle East

Saipem recently undertook a trunkline pipeline project in the Arabian Gulf for which the as-laid survey was accomplished using the FlatFish UID. This project offered the opportunity to test the concept of using a UID in *untethered ROV mode*. This type of survey includes the requirement to confirm conformance to the design alignment as well as check for and measure any free-spans and confirm integrity of any pipeline/cable crossings. Normally this is performed using a conventional ROV from a dedicated vessel.

The *untethered ROV mode* with WiFi telemetry proved the concept and gave future confidence in the ability to eliminate or reduce the need for a full-time dedicated support vessel to track along the pipeline. Instead, the support vessel could be freed to undertake other pipelay related tasks or eliminated entirely, allowing the UID to be controlled from the pipelay vessel kilometres from the survey location.

Some lessons were learned and some innovative new potential approaches have been developed as a result of this first use case, but the overall result was very positive and validated the concept as a viable and superior alternative for shallow water pipeline survey compared to a conventional ROV.

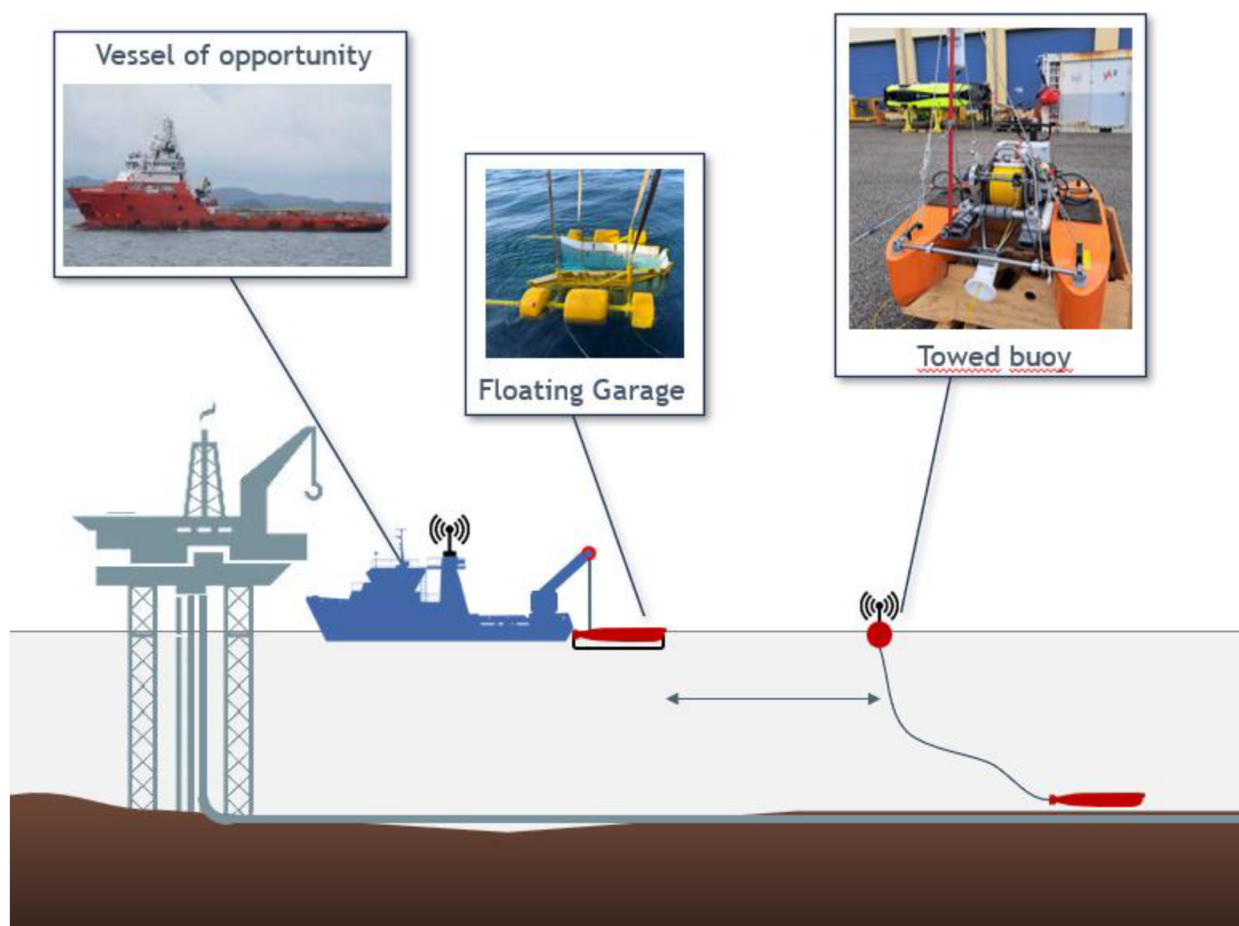


Figure 4—Subsea drones operated in shallow water application through the wi-fi buoy

Acquired data

The as-laid survey acquisition was performed using the UID, covering a total of 97 km of pipeline. Data acquisition was carried out in accordance with industry standards, utilizing the vehicle's integrated survey equipment. This included a high-resolution laser system for precise free-span detection. The CathX camera system delivered exceptionally detailed imagery, suitable for 3D photogrammetry and point cloud generation if required. The quality of data obtained significantly surpasses that of conventional ROV cameras, offering advanced visual clarity and survey data.

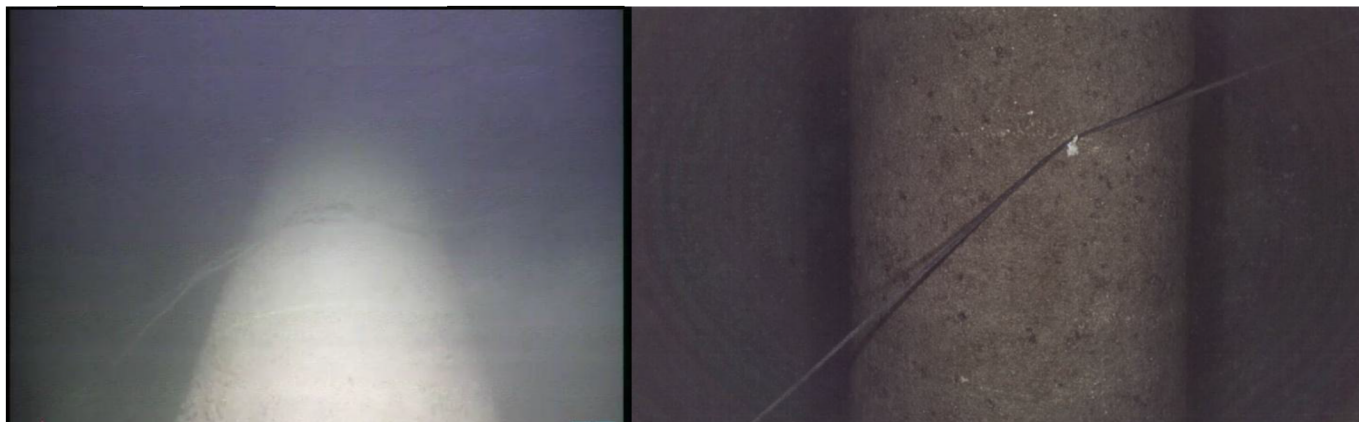


Figure 5—Pipeline visual inspection comparing conventional ROV acquired data (on the left) and UID high definition data (on the right).

During this first-use case, the following observations were made:

- **Speed of survey**

It was noted that the UID was able to cover significantly more pipeline per day than possible with a conventional ROV. Typical observed survey speeds were in the order of 2500m/hr. Despite the need for long (6hrs) recharging on deck, it is estimated that the UID could achieve around two to three times the daily coverage of an ROV, or even better as far as the battery technology evolves and the technology maturity progresses. Additionally, the UID exhibited good tracking and position keeping despite maximum current of around 1 m/sec.

- **Quality of Data and Video**

As illustrated in Figure 4 above, the video quality obtained from the UID is very high resolution, thus greater detection and far more accurate assessment of any observed anomalies.

- **UID remote control and response via the Towed Wi-Fi Communications Buoy**

Initially, until sufficient confidence in the modified control system was achieved, the UID was operated in ROV mode with a light communication tether connected directly to the support vessel. Eventually, the UID was deployed with the WiFi-linked Communications Buoy (Fig. 3) and operated in *Untethered ROV mode*.

In untethered ROV mode control of the UID movement as well as live video streaming and MBES data were all transferred between the support vessel control room and the UID wirelessly. Quality of the video and data was observed to be unaffected by the telemetry link and control of the UID was similarly unaffected. It is noted however, that should there have been a loss of telemetry, the UID was programmed to stop itself and wait for comms restoration or eventually move to surface waiting for an intervention.

- **Launch & Recovery**

It had been the intention to launch and recover the UID using the *Floating Garage* (Fig.3), however this method was not designed with joint deployment with the communications buoy in mind and proved to be problematic. Ultimately, launch and recovery was achieved by directly lowering with the vessel crane and a quick release hook. Alternative solutions have been identified and are currently being studied.

- **Fishing net / obstacle avoidance**

One of the traditional hazards of operating an ROV or towed fish in the Arabian Gulf and other areas of the world, is the prevalence of fishing nets and lines. On one occasion, a fishing line became entangled in the UID thrusters, requiring surface intervention to clear.

The UID is fitted with acoustic and visual obstacle detection and automated avoidance. Even in ROV mode, the UID will automatically stop when confronted with an obstacle and seek a route around it. However, fishing nets and lines can be difficult to detect and snagging can not be fully eliminated. To this end, improved obstacle detection algorithms have now been developed which should further reduce the likelihood of getting snagged.

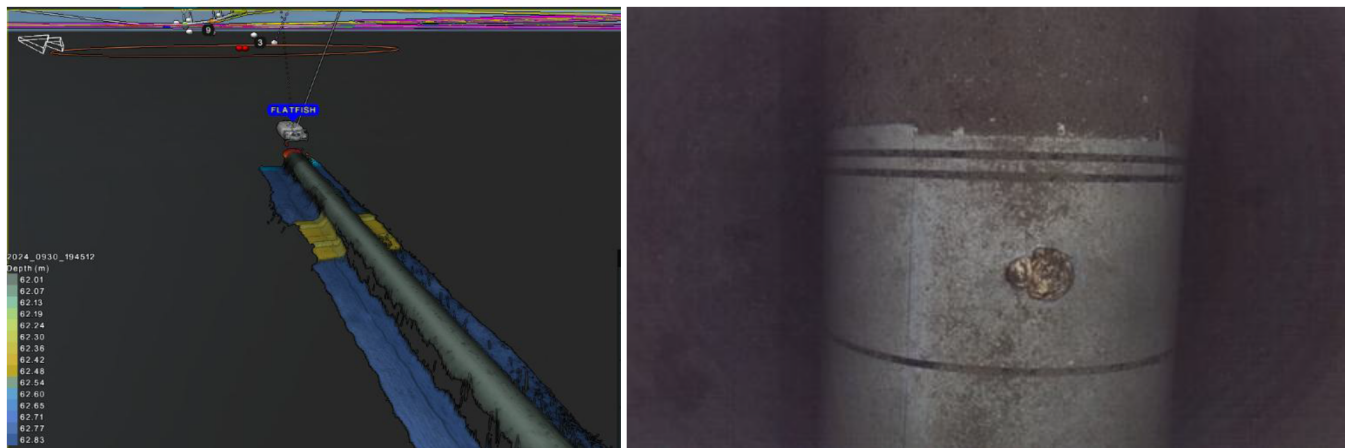


Figure 6—UID acoustic data (on the left) and high definition detail of an anomaly with visual inspection (on the right).

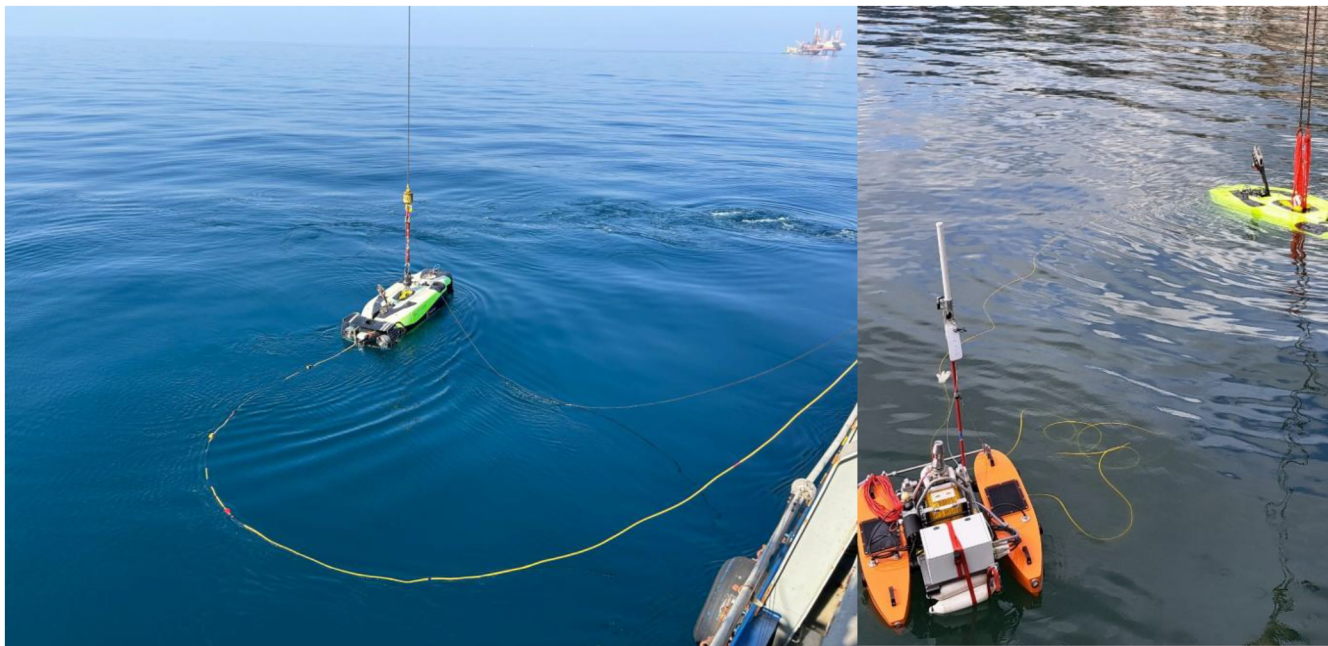


Figure 7—UID in conventional tethered ROV mode (on the left) and connected to the Wifi Buoy for untethered mode (on the right).

Performance comparison between ROV mode UID and Conventional ROV

The following observations and conclusions are made when comparing the adapted UID with a conventional ROV method:

Criteria	ROV mode UID (<i>Saipem FlatFish</i>)	Conventional ROV
Productivity	- Nominally around 15-20 km per day allowing for deployment, repositioning	- Nominal ROV pipeline tracking around 8 - 12 km per day (allowing for 18-20 hrs/day in-water).

Criteria	ROV mode UID (<i>Saipem FlatFish</i>)	Conventional ROV
	and charging (12-16 hrs/day in-water) and depending on environmental conditions.	
Operational Downtime	- Nominal Operational downtime (time on deck) in steady-state pipeline inspection operations is around 6-8 hours per day - Recharge time, including any data upload / mission downloads takes typically 6 hours. - The UID can nominally operate for 12-16 hours between recharges.	- Normally 2-4 hours per day time on deck for maintenance / reconfiguration.
Operational Flexibility	- Once deployed the support vessel is free to move off and undertake parallel operations such as ROV or Diving inspection of a platform. - The UID has full 6 degrees of freedom maneuverability and ability to hover and station-keep.	- Vessel needs to permanently follow the ROV in close proximity due to tether length constrains.
Near-Platform Operations	- In untethered ROV mode the vessel is not required to be in close quarters to platforms.	- WROV is restricted by need to have vessel in close quarters with the attendant collision risk.
Crew Fatigue Risk	- In ROV mode, much of the navigation, tracking and altitude control is automated, greatly reducing Pilot fatigue.	- Pilots must continuously monitor and actively control the ROV
Safety	- In untethered ROV mode the UID allows the vessel to move well away from the pipeline (about 1 km). This eliminates H2S risk for vessel crew in the event of a leak. - Similarly, platform collision risk is eliminated because the vessel can remain outside the 500m zone and well clear of any platforms while the UID inspects pipelines to the riser bends.	- Vessel needs to permanently follow the ROV in close proximity due to tether length constrains.
Sensors for Pipeline Survey & Inspection	The UID has: - Stereo Cath-X UHD cameras and lasers recording which allows mm accurate geo-referenced point-cloud 3D Photogrammetry which can be used for metrology and 3D modelling / rendering. - The rendered imagery can be combined with the MBES scans to provide highly detailed composite imagery. - Built-in leak detection capability - Contactless CP measurement skid available.	Normally ROV fitted with: - UHD camera system for video recording - MBES - SSS
Seabed survey	- The UID can undertake a seabed survey using MBES and UHD stereo video operating in either ROV mode, Supervised AUV mode or AUV mode. - In the absence of any known line feature or structure, autonomous INS navigation will be based on pre-programmed grid coordinates. - In very shallow waters (1.5m to 3m) and in flat calm weather, the UID can operate at the surface to acquire data.	- Technically possible but requires accurate depth (altitude off bottom) control. - Cannot operate in very shallow waters (<around 5-10m)

Conclusion

This paper has presented the experience and learnings from the use of a UID designed to be normally fully autonomous in a novel way by adapting it for use in an untethered, direct-controlled ROV mode for pipeline surveys in shallow waters. In this mode, it was found to offer significant advantages over a conventional ROV, especially when conducting as-laid pipeline surveys. The specific advantages observed were:

- Increased productivity in terms of kilometres per day,
- Increased operational flexibility in terms of freeing up the support vessel for other parallel activities,
- Increased cost effectiveness due to the productivity and operational flexibility mentioned above,
- Decreased risk of collision when surveying near platforms and other structures,
- Reduced crew fatigue because of the semi-automatic feature detection and tracking,
- Enhanced video and data quality allowing 3D modelling and/or metrology if required,
- Ability to operate in shallow water closer to the beach than an ROV can reach.

Certain drawbacks and potential improvements were also noted:

- Time between charges varied with environmental loading (greater battery draw). An upgrade the battery is already planned to significantly enhance capacity such that time between charges will be almost doubled. Recharging takes around six hours (before battery increase). The planned battery upgrade will also allow for rapid charging and will decrease this time.
- Fishing net entanglement is still a risk. Automatic obstacle avoidance using AI and machine learning is already planned to be improved with the next round of software upgrades.

Recommended Applications / Use-cases

Apart from the demonstrated benefits of using the UID in ROV mode for conducting as-laid pipeline surveys, the following additional tasks would probably benefit as well:

- Pipelay support – startup / laydown / touchdown / crossing monitoring launching the UID directly from the pipe lay vessel avoiding the mobilization of a dedicated support vessel, generating a significant cost reduction
- Pre-lay survey – allowing seabed survey of extended areas by autonomous grid paths navigation
- As-Built survey
- Investigation of potentially leaking pipelines – especially where risk of H₂S poisoning would preclude divers or the positioning of the support vessel close to the leak location.
- Baseline / recurrent IMR Inspection
- Known/suspected pipeline anomaly detection and evaluation

Nomenclature

AI	Artificial Intelligence
AUV	Autonomous Underwater Vehicle
CP	Cathodic Protection
CVI	Close Visual Inspection
DP	Dynamic positioning
GVI	General Visual Inspection
H ₂ S	Hydrogen Sulfide
IMR	Inspection, Maintenance and Repair
LARS	Launch & Recovery System
MBES	Multi Beam Echo Sounder
MSV	Multi Support Vessel
ROV	Remotely Operated Vehicle

SBP	Sub Bottom Profiler
SSS	Side Scan Sonar
TDP	Touch Down Point
UID	Underwater Intervention/Inspection Drone
USBL	Ultra Short Baseline
USV	Unmanned Surface Vessel
UHD	Ultra High Definition

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